

I recap

II Broad Overview

Experiment
Sample space $S \ni s$ outcomes
 $p: S \rightarrow \mathbb{R}, p(s) \in [0, 1], \sum_{s \in S} p(s) = 1$.

Complete description of result.

Random Variables

$X: S \rightarrow \mathbb{R}$

model numerical "observables" of an experiment.

Average

$$EX = \sum_{s \in S} p(s) X(s)$$

<domain, useful in proofs>

$$= \sum_{r \in \mathbb{R}} r \underbrace{P(X=r)}_{\text{distribution of } X}$$

<range, useful in applications>

Algebra: $X+Y, cX, XY$

* Linearity: $E(X+Y) = EX + EY$

Variance: $V(X) = E(X - EX)^2$

Average squared deviation from EX

* If $X_1, \dots, X_n$ indep then $V(X_1 + \dots + X_n) = V(X_1) + \dots + V(X_n)$

Events $E, F \subseteq S$

crisis $\rightarrow$ $P(E) = \sum_{s \in E} p(s)$ outcomes

$$P(E|F) = P(E \cap F) / P(F)$$



Independent: $P(E|F) = P(E)$ (P(F) > 0)

$$P(F|E) = P(F)$$

(P(E) > 0)

$$P(E \cap F) = P(E)P(F)$$

Allows decomposition of complicated IP into simpler.

Bayes: $P(E|F) = P(F|E) \frac{P(E)}{P(F)}$

Total Prob: $P(E) = P(E|F)P(F) + P(E|\bar{F})P(\bar{F})$

Set th.: $P(E \cup F) = P(E) + P(F) - P(E \cap F)$

Disjoint: $E \cap F = \emptyset$

M: $P(Y \geq E) \leq \frac{EY}{E}, Y \geq 0$

C: $IP(Y \leq \frac{EY + V(Y)}{2}) \geq \frac{1}{2}$

Markov & Chebyshev

Markov: If $Y \geq 0$, then

$$Y(s) \geq 0 \quad \forall s \in S$$

$$P(Y \geq t) \leq \frac{EY}{t}$$

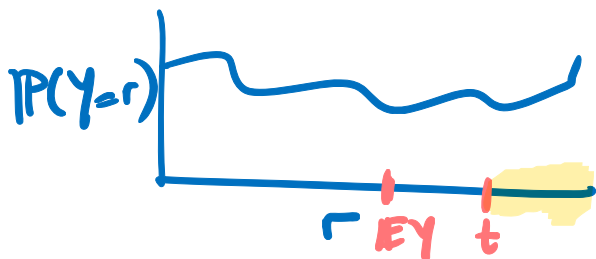
$$t = c EY$$

$$P(Y \geq c EY) \leq \frac{1}{c}$$

$$c=2: P(Y \text{ is twice as big as } EY) \leq \frac{1}{2}$$

$$c=3: P(\text{thrice } \dots) \leq \frac{1}{3}$$

"the prob. that a nonnegative r.v. is bigger than its average is small".



one sided bound.

$$Z = (Y - EY)^2$$

Chebyshev:

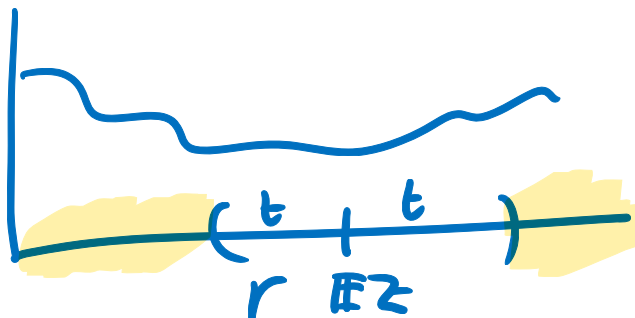
$$P(|Z - EZ| \geq t) \leq \frac{V(Z)}{t^2}$$

$$SD(Z) = \sqrt{V(X)}$$

$$t = c SD(Z)$$

$$P(|Z - EZ| \geq c SD(Z)) \leq \frac{1}{c^2}$$

two-sided bound.



Examples:

Bernoulli(q)

Single coin flip
w/ bias q .

$$\text{distribution} \quad X = \begin{cases} 1 & \text{w.p. } q \\ 0 & \text{w.p. } 1-q \end{cases}$$

$$\mathbb{E} \quad q$$

$$\mathbb{V}(\cdot) \quad q(1-q)$$

$$Y = X_1 + \dots + X_n, X_i \sim \text{Bernoulli}(q)$$

linearly

indep.

Binomial(n, q)

n independent
coin flips w/ bias q .

$$Y = \begin{cases} k & q^k (1-q)^{n-k} \binom{n}{k} \\ \text{for } k=0, 1, \dots, n \end{cases}$$

$$qn$$

$$q(1-q)n$$

Geometric(q)

of independent
Bernoulli(q) trials
until success.

$$Z = \begin{cases} m & (1-q)^{m-1} q \\ m=1, 2, 3, \dots \end{cases}$$

$$\frac{1}{q}$$

$$—$$

— Throw m balls in n
bins, independently.

$$E_{ij} = \{ \text{Bin } i \text{ contains ball } j \} \quad \begin{matrix} i=1 \dots n \\ j=1 \dots m \end{matrix}$$

$$1_{E_{ij}} \sim \text{Bernoulli}\left(\frac{1}{n}\right)$$

$$X_i = \# \text{ of balls in bin } i \quad i=1 \dots n$$
$$\sim \text{Binomial}\left(m, \frac{1}{n}\right).$$

f.v. focus on one feature
of the experiment.

— Coupon Collector: Roll an n -sided die
 until you see m distinct numbers.

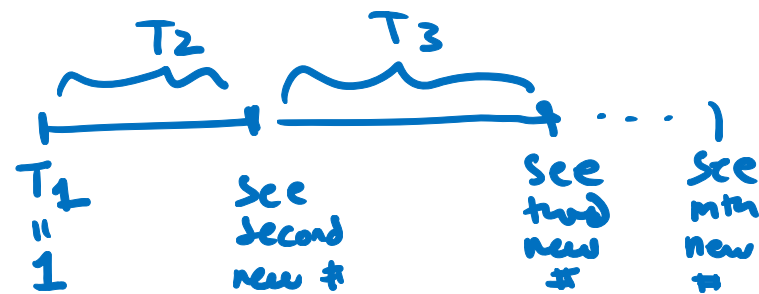
T = total # of rolls.

$\text{Geom}(1) \sim T_1 = 1$

$\text{Geom}(\frac{n-1}{n}) \sim T_2 = \# \text{ of rolls after } T_1 \text{ to see second new \#}$

$\text{Geom}(\frac{n-2}{n}) \sim T_3$

$\text{Geom}(\frac{n-(i-1)}{n}) \sim T_i = \# \text{ of rolls after seeing } i-1 \text{ distinct numbers to see } i\text{th new \#}.$



ex: $n=4, m=3.$

$s = \underbrace{2}_T \underbrace{2}_T \underbrace{1}_T \underbrace{2}_T \underbrace{1}_T \underbrace{4}_T$

$T_1(s)=1, T_2(s)=2, T_3(s)=3.$

$$T = T_1 + T_2 + \dots + T_m$$

Remark: If 5 cards are drawn from a deck of 52:

$$S_1 = \{s: 5\text{-permutations of } [52]\} \quad S_2 = \{s: 5\text{-combinations of } [52]\}$$

$\uparrow$ order matters. $\uparrow$ order doesn't matter.

$$p(s) = \frac{1}{P(52, 5)}$$

ok if all events
don't depend on order of cards.

$$p(s) = \frac{1}{\binom{52}{5}}$$

In general, use labeled arrangements where order matters
in defining sample spaces.:

ex: Flip 10 fair coins.

$$S_1 = \{0, 1\}^{10}$$

(Coins are distinguishable)

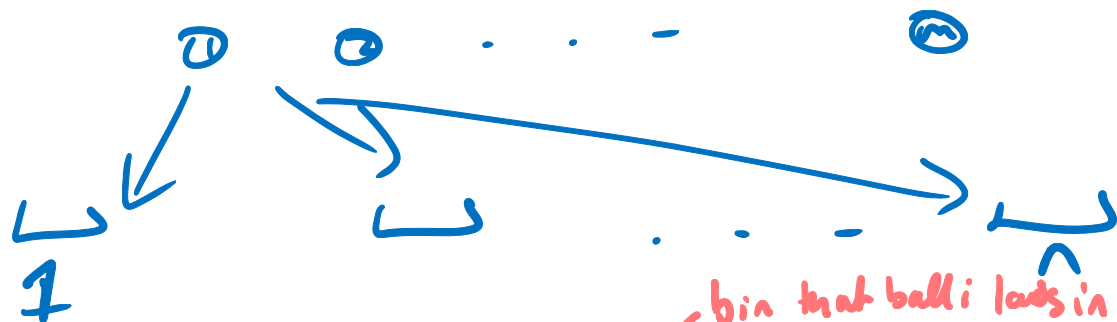
$$p(s) = \frac{1}{2^{10}}$$

Tricky (avoid):

$$S_2 = \{ \# \text{ of arrangements of } 10 \text{ 0/1's where order doesn't matter} \}$$
$$= \{0, 1, 2, \dots, 10\}.$$

$$p(k) = \frac{\binom{10}{k}}{2^{10}}$$

Remark: Throw m balls in n bins ^{independently} uniformly at random.
 $E = \{ \text{first bin is empty} \}$. What is $P(E)$?



Counting:

$$S = \{ (b_1, \dots, b_m) : b_i \in \{1, \dots, n\} \}$$

$$E = \{ \underline{b} \in S : b_i \neq 1 \forall i \} \leftarrow \text{ways to put } m \text{ balls in bins } 2 \dots n.$$

$$P(E) = \frac{|E|}{|S|} = \frac{(n-1)^m}{n^m}.$$

↓
Count configs involving all bins.

Probability: $P(E) = P(E_1 \cap E_2 \cap E_3 \dots E_m)$ where $E_i = \{ \text{ith ball} \neq \text{bin } 1 \}$
 $= \{ b_i \neq 1 \}$.

$$\stackrel{\text{indep}}{=} P(E_1) P(E_2) \dots P(E_m)$$

$$= \frac{n-1}{n} \cdot \frac{n-1}{n} \dots \frac{n-1}{n} = \left(\frac{n-1}{n} \right)^m.$$

↪ no counting, just look at one bin at a time.

Tips :

- (1) Always write down the experiment explicitly.
- (2) Always name events, E, F , etc.
- (3) Express the given data in terms of $P(E)$, $P(E|F)$, etc.
- (4) Stick to the formal rules.

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- Cheat sheet: 2-sided handwriter
 - Much less time pressure on final.